

MORRIS, IL, USA

WMO: 720137

Lat: 41.425N

Lon: 88.419W

Elev: 178

StdP: 99.20

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04867

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-17.0	-23.8	0.4	-19.5	-21.0	0.6	-16.1	13.3	-0.1	12.0	-3.7	2.9	290	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.5	22.9	31.2	22.6	29.8	22.1	25.4	29.8	24.5	28.8	23.6	27.7	4.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.3	27.8	23.0	18.1	26.9	22.4	17.4	26.3	78.7	29.8	74.8	28.9	71.4	27.9	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.9	8.5	DB	-23.5	34.7	3.7	1.2	-26.2	35.6	-28.4	36.3	-30.4	37.0	-33.1	37.9	(4)
(5)			WB	-24.1	27.2	3.6	0.7	-26.7	27.7	-28.9	28.1	-30.9	28.6	-33.6	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-4.3	-3.2	4.0	10.0	16.6	21.5	22.8	21.7	18.9	11.7	4.6	-1.6	(6)
(7)		DBStd	10.97	6.69	6.67	6.00	5.07	5.09	3.34	2.95	2.80	4.13	5.27	5.24	6.01	(7)
(8)		HDD10.0	1660	443	371	204	61	6	0	0	0	1	41	174	360	(8)
(9)		HDD18.3	3398	701	603	445	252	96	9	2	5	42	215	411	617	(9)
(10)		CDD10.0	1774	1	1	18	62	209	346	398	363	267	96	13	1	(10)
(11)		CDD18.3	471	0	0	1	3	42	104	142	110	59	11	0	0	(11)
(12)		CDH23.3	4254	0	0	12	51	432	938	1295	883	537	107	0	0	(12)
(13)		CDH26.7	1322	0	0	1	6	146	313	423	237	168	26	0	0	(13)
(14)	Wind	WSAvg	3.3	4.3	4.0	3.9	4.1	3.5	2.8	1.9	1.6	2.1	3.1	3.8	3.9	(14)
(15)	Precipitation	PrecAvg	928	51	41	59	93	112	114	107	99	78	72	65	50	(15)
(16)		PrecMax	1183	126	140	141	185	188	293	212	288	295	218	157	139	(16)
(17)		PrecMin	607	10	11	12	36	25	25	7	16	17	10	7	9	(17)
(18)		PrecStd	138	30	26	30	41	42	61	55	62	59	48	39	29	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	16.7	25.3	27.5	32.8	33.6	34.8	32.9	33.0	29.9	22.0	15.7	(19)
(20)			MCWB	12.2	12.3	17.5	16.7	20.0	22.7	25.0	23.1	22.8	20.9	13.7	13.5	(20)
(21)		2%	DB	9.9	12.1	19.6	24.8	30.4	32.2	32.2	31.0	31.1	26.3	17.7	11.9	(21)
(22)			MCWB	8.5	9.2	14.4	15.5	19.6	22.5	24.5	23.3	22.5	17.4	12.9	9.9	(22)
(23)		5%	DB	6.7	7.7	16.3	22.3	27.9	30.2	30.4	29.2	28.5	23.1	15.4	8.9	(23)
(24)			MCWB	5.3	5.5	12.4	14.2	18.7	21.6	23.5	22.7	21.2	16.3	11.7	7.7	(24)
(25)		10%	DB	4.0	4.9	13.4	19.7	25.7	28.0	29.0	27.8	26.7	20.7	12.6	6.3	(25)
(26)			MCWB	2.8	3.3	10.0	13.3	18.1	20.6	22.7	21.9	19.5	16.0	9.5	4.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.7	13.5	17.8	18.8	22.8	26.0	27.1	25.9	25.1	22.2	16.5	14.3	(27)
(28)			MCD B	13.2	15.1	22.9	24.8	28.0	30.9	31.3	30.1	30.1	27.6	19.4	15.6	(28)
(29)		2%	WB	8.7	9.4	15.5	16.9	21.5	24.3	25.7	24.8	23.8	20.2	14.1	10.2	(29)
(30)			MCD B	9.6	11.6	18.9	21.9	26.9	28.8	29.9	28.9	28.5	24.0	16.7	10.8	(30)
(31)		5%	WB	5.4	6.0	13.3	15.5	20.4	23.2	24.8	23.9	22.4	18.5	12.0	7.6	(31)
(32)			MCD B	6.8	7.8	15.9	20.6	25.7	27.7	29.0	27.7	26.3	21.4	15.0	8.4	(32)
(33)		10%	WB	2.9	3.1	10.1	13.9	19.3	22.3	23.8	23.0	21.1	16.6	10.1	5.1	(33)
(34)			MCD B	3.9	4.5	13.0	18.8	24.0	26.6	27.9	26.4	24.6	19.5	12.8	6.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.9	8.4	9.4	11.7	11.8	11.1	10.9	10.8	12.2	11.3	9.3	7.4	(35)
(36)			MCDBR	9.8	11.4	13.7	15.8	14.6	12.9	11.6	12.1	13.8	14.9	12.9	10.2	(36)
(37)			MCWBR	8.4	8.8	8.9	8.6	6.6	5.8	5.6	6.0	6.2	7.5	8.7	8.7	(37)
(38)		5% WB	MCD BR	9.6	11.2	12.2	14.3	12.1	10.5	10.5	10.1	11.3	11.6	11.2	9.5	(38)
(39)			MCWBR	8.5	9.2	8.7	9.0	6.7	5.7	6.2	5.8	5.9	7.5	8.5	8.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.311	0.346	0.400	0.430	0.437	0.453	0.442	0.412	0.355	0.311	0.292	(40)	
(41)		taud	2.430	2.381	2.381	2.240	2.211	2.234	2.181	2.203	2.258	2.419	2.492	2.451	(41)	
(42)		Ebn at Noon	864	898	902	870	849	842	823	820	819	835	836	832	(42)	
(43)		Edh at Noon	78	96	108	133	141	138	144	137	121	92	73	71	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.66	2.55	3.59	4.66	5.32	6.05	6.05	5.34	4.49	2.98	1.92	1.41	(44)	
(45)		RadStd	0.16	0.33	0.27	0.38	0.43	0.44	0.34	0.31	0.39	0.34	0.17	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page